

BioCoLoop: PUBLIC-HARNESS COMPARISON

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Public frameworks use their task-adapted native experimental controllers and the common biological fitting interface. AI-Scientist-v2 and AI-Researcher access laboratory 0, whereas BioCoLoop accesses ten laboratories. All rows use Qwen2.5-7B-Instruct, the same twelve-design library, at most six candidate proposals and the same held-out scorer. Scored cells pass independent recomputation from saved predictions; failure tokens record terminal controller outcomes rather than predictive scores.

Model / method	DTI		Proteomics		Cell perturbation		
	BindingDB		PTPC		VCC		Tahoe
	AUROC $\uparrow$		AP $\uparrow$		Top-1 $\uparrow$		Top-1 $\uparrow$
Fixed model (1 lab)	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00		
Qwen direct (1 lab)	89.89 $\pm$ 1.02	20.13 $\pm$ 0.07	21.09 $\pm$ 4.38	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21		
AI-Scientist-v2 (1 lab)	89.73 $\pm$ 1.13 (2/3)	20.18 $\pm$ 0.43	18.61 $\pm$ 2.78	14.12 $\pm$ 0.04	17.91 $\pm$ 6.33 (2/3)		
AI-Researcher (1 lab)	86.09 $\pm$ 0.88	20.12 $\pm$ 0.05	19.70 $\pm$ 2.87 (2/3)	21.73 $\pm$ 6.63	17.41 $\pm$ 6.21		
BioCoLoop (10 labs)	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22	31.54 $\pm$ 0.08	<u>22.25</u> $\pm$ 0.45	22.39 $\pm$ 2.59		

Table 1: Three-seed comparison (mean $\pm$ sample SD; seeds 42–44). Public methods use laboratory 0; BioCoLoop uses ten. All rows share the Qwen2.5-7B-Instruct backend, 12-design library, six-candidate cap, 100-round fits and held-out scorer. Scores are $\times 100$; bold/underline mark the best/second-best complete result. ($n/3$) reports incomplete cells.

Outcome. The flat table reports the three-seed mean and sample SD for every complete cell, and preserves the completion count for public-controller failures. Bold and underline identify the best and second-best complete values within each endpoint; no incomplete controller cell is ranked.